

# Data Advocacy for Visibility of Home Care Workers

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## ABSTRACT

Technology solutions have been proposed to address the “care gap” that the United States has been facing due to the increasing demand for home and community based services that far exceeds the supply of caregivers. However, these technology solutions do not fully address the crux of the crisis—the poor working conditions of home care workers (HCWs). Rather, these technology solutions further burden, invisibilize, and surveil workers. My research challenges this dynamic by proposing the design of technology that can be used to advocate for the workers, amplifying their voices and highlighting their work. In this paper, I describe the three phases of my research where I: (1) examine the relationship between visibility and technology, (2) explore different design approaches to see how technology could complicate questions of visibility, and (3) build and deploy a system to evaluate how data could be used for advocacy. Helping workers collect and share data could build awareness and solidarity, expose employer violations and uncompensated work, and challenge existing narratives of care.

## CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

## KEYWORDS

frontline health, worker advocacy, collective action, data activism

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## 1 BACKGROUND AND MOTIVATION

Starting this January, Section 120006(a) of the 21st Century Cures Act mandated that all states were required to use electronic visit verification (EVV) smartphone applications for tracking all in-home care visits funded by Medicaid. EVV was proposed as an approach to America’s caregiving crisis, facilitating the more efficient use of Medicare funds by streamlining service delivery and management of a growing labor force by focusing on “fraud, waste, and abuse” [14]. However, like many other top-down imposed workplace technologies, EVV created more inefficiencies than it saved, with a lot of the costs falling onto the shoulders of the workers and

“further[s] entrenchment of unequal power relations” [11]. As most HCWs are older adults themselves, they reported having to spend extra time learning how to use the applications and dealing with the issues they presented. Additionally, because the technologies are geared towards compliance, the checkboxes HCWs fill out in the applications do not account for the emotional or articulation work they perform [25]. Finally, invasive features like location-tracking and photo verification mean that both HCWs and their vulnerable clients feel restricted in their agency [14].

The reason these technologies fall short of impacting the caregiving crisis is that they do not address a core problem—the poor working conditions of home care workers (HCWs). One of the main drivers of the “care gap” is the high turnover rate of HCWs, averaging at 65% in 2020. Even though HCWs provide essential care in the homes of older adults and people with disabilities and empower them to thrive in their communities, HCWs are themselves precarious workers. Due to low wages and inconsistent hours, HCWs face high levels of financial insecurity, with more than half receiving public assistance. Moreover, they face high levels of physical and emotional burden [21]. This results in home care work being seen as undesirable, “jobs of last resort” left to women of color and immigrants who have no other choices [9].

Technology and data have the potential to address these issues by highlighting worker voice and invisible work. Technology and data have helped aggregate inputs from othered voices, building a shared narrative across geographically alienated individuals (e.g., online storytelling [6]) or holding institutions accountable through external accounting (e.g., self-survey [19]). Worker-centered technologies [8] (e.g., Turkopticon [12], Shipt Calculator [4]) have reduced information asymmetry and empowered workers to shift power dynamics in their favor. More specifically, prior literature demonstrates how technology and data can serve as a mobilizing structure and a new part of activists’ repertoire of contention, enabling them to act on injustice and hold oppressors accountable through a “reversal of the Foucauldian panopticon” [13].

However, the literature also raises questions around what visibility means and how visibility should be achieved. There are many levels of visibility across different stakeholders. Building on the social ecological model [23], we see that visibility could be on the level of individuals (i.e., workers understanding the value of their own contributions and being empowered to advocate for themselves), the interpersonal (i.e., being respected by their clients and the clients’ families and recognized for their contributions), organizational (i.e., feeling supported by employers for the issues that they run into and assurance that they will not happen again), community (i.e., sharing knowledge, data, and solidarity with other workers and worker advocates), and policy (i.e., being acknowledged in workplace legislation and having funding that reflects their levels of contribution.). My research explores the question of what level of

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visibility technology could and should impact. The second question is around mitigating the potential externalities that may occur in the pursuit of visibility. There are questions around how care could be operationalized (i.e., some elements of care like “humor” are difficult to codify [2]), how visibility could be balanced with surveillance (i.e., whether visibility leads more towards recognition and protection or additional scrutiny [22]), and how to relieve worker burden (i.e., developing tools could help empower workers to make change, but might put the onus on them to do so [20]). My research builds on these questions by exploring multimodal tracking [18], participatory privacy [17], and new technological affordances [28].

## 2 APPROACH AND METHODS

My research aims to build on this literature to understand **How might we use technology and data to advocate for more visibility for home care workers?** In order to explore this question, I first examined what visibility meant to workers and how technology could not only enable but also subvert this visibility. This led to the first contribution, which was a framework of invisible work and technology in the home care context. Then, based on this theoretical understanding, I explored how different designs of technology could complicate these questions of visibility. From there, I made the contribution of having guidelines for designing for visibility. Building on these guidelines, I plan to build and deploy a pilot pipeline to see how these theories and ideas could be combined. From there, I will have a better understanding of how the design principles for visibility play out in a real-world deployment and how the data that is collected could be used for advocacy.

### 2.1 Examining Visibility and Technology

The first phase of my research (published as [16]) builds on prior work that has documented examples of invisible work in frontline health (e.g., [27]) and how technology enables invisible work (e.g., [26]) by examining invisible work and technology in the home care context. In order to do this, I conducted interviews with 13 workers recruited through partnerships with 1199SEIU, a union in New York City, coding these interviews and then applying thematic analysis [3]. I found many examples of how HCWs had to “go beyond and beyond” but were not recognized for it. As one of the participant summarized:

*“You need a big heart. You need to be able to be a psychiatrist, their secretary, their grocery store. That’s what you got to be. You got to be more than a home health aide. So that’s what I’m saying. That we need more money. We need more incentive. We need more.” (P1)*

In addition, this research found that understanding the relationship between technology and invisible work could help us change that dynamic. I found how technology, as it is currently designed, enabled invisible work (i.e., developing the “sociotechnical” mechanism of invisibility [10]). However, if technology design were to take into account both task-based inefficiencies and emotional labor, it might be able to improve worker experiences. Doing so would require careful consideration of tensions around autonomy, actionability, and impact.

### 2.2 Exploring Designs for Visibility

The next phase was figuring out how we might make these notions of technology and visibility tangible in design (submitted as [15]). I spoke to Healthcare Workers Rising, the Upstate New York offshoot of 1199SEIU, and learned that one of the problems that their workers were facing was that not being properly compensated for their work and in that sense, their work was being rendered invisible. Building on examples in literature of how technology has enabled this invisibility in the form of wage theft [24] and how it could potentially serve as solutions [7], I explored wage theft in the unique context of home care. First, I conducted individual interviews with experts and workers to understand why wage theft happens, what HCWs can do about it, and what challenges there are in the process. Then, based on these interviews, I created design provocations [1] for identifying and acting on wage theft that served as a starting point for focus group conversations about technological solutions. Through these provocations, I was able to elicit different perspectives on how employer-driven and worker-driven technology could be used to address individual worker issues (e.g., having an easier way to raise and track the progress of issues) and enable collective action (e.g., sharing knowledge and aggregating their information). Moreover, I raised tensions around potential technological futures, including questions around the worker burden of the invisible work of tracking and organizing as well as handling the sensitivity of the collected data.

### 2.3 Developing Visibility to Advocacy

Based on the ideas and tensions raised by previous steps, I want to explore what it would look like for a system to be able to help workers collect and share data that would not only make visible violations like wage theft but also the context around them. Therefore, in the next phase of my research, I plan to pilot a short-term evaluation of a potential data pipeline that would be able to take numbers and stories from HCWs and produce outputs that can further advocacy aims. Throughout this process, I will apply a community-engaged, iterative approach drawing from design justice principles [5] to allow me to better understand worker and worker advocate perspectives and context. I will be continuing to work with Healthcare Workers Rising as well as with existing statewide campaigns such as Fair Pay for Home Care and Securing Wages Earned Against Theft.

**Designing & Building the System.** One of the main goals of this research is to be able to address questions around how we might build a system that balances impact, privacy, and burden. More specifically, we are interested in seeing how to best balance the desired use of the data collectively by the advocates might conflict with the challenges and incentives presented to individual workers. We hope to do so by collecting both number and stories to build comprehensive narratives, eliciting privacy preferences and controls from workers, and leveraging new technological affordances (e.g., passive location tracking) and stakeholders to reduce the burden on workers. In order to address these questions, I first will have 3-5 focus groups with 3-5 worker advocates to figure out what types of data we could collect and how that data could be used. Then, I will develop medium-fidelity prototypes to present to 3-5 focus groups of 3-5 workers to understand privacy and burden

perspectives of how that information could be collected. Next, I would develop the system based on the designs and feedback and evaluate this system through a short-term (1-2 week) deployment with a small (5-10) group of workers. Finally, I would be able to make any changes to the application to account for any issues that arise.

**Analyzing & Presenting the Data.** Another goal of the system is to see how data could be collected and shared for advocacy. Specifically, I would be interested in seeing if the system would allow worker advocates to be able to identify issues and collect evidence that support their efforts. In order to do this, I will deploy the system for 2-3 months with 12-15 workers, maintaining monthly check-ins to make any necessary changes and build group connections. I will also be paying the workers to use the application because the aim of the project is to evaluate the data pipeline rather than necessarily have a fully-fledged, sustainable application. After the data collection period, I will analyze the data and present these insights to focus groups with workers, advocates, and policymakers. Ideally, these outputs would be able to be used for the advocacy period before the New York State budget hearings in early spring of next year.

### 3 OUTCOMES AND CONTRIBUTIONS

The main contributions of my research include: (1) a deeper understanding of the relationship between invisible work and technology, (2) design principles for handling tensions that come from visibility, and (3) the evaluation of an end-to-end system that would help workers, worker advocates, and policymakers better understand and advocate for care. In terms of which level of visibility to target, my research explores the perspectives of different stakeholders and what conflicts may arise. Moreover, in terms of mitigating externalities of visibility, I investigate how to best balance these conflicts in the design and deployment of a pilot pipeline. My research will contribute to theoretical concepts of care, visibility, and impact by seeing how technology that shifts power to workers could lead to structural change. Beyond HCWs, I explore questions around quantifying care by using technology and data as a means of making it more visible. Additionally, I examine how to do so in a way that honors the voices of marginalized workers. Finally, by grounding the project in a real-world deployment, I investigate the potential futures and impacts on policy and public opinion.

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